

IEOR 4004 Assignment 2

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Question 1

Part 1: Formulate and solve a linear program to minimize the dispatcher labor costs

Problem Summary: Three airports in New York City operate 24 hours a day, 7 days a week. The total number of air traffic controllers needed during each 4-hour interval is given in Table 1. Controllers work either 8-hour or 12-hour shifts. 8-hour shifts start every 4 hours (00:00, 04:00, 08:00, 12:00, 16:00, 20:00). 12-hour shifts start only at 00:00, 08:00, 12:00, or 20:00. Cost: \$93/hour (8h shifts), \$100.8/hour (12h shifts).

Table 1 – Demand

Time Interval	Controllers Needed
12 a.m. to 4 a.m.	8
4 a.m. to 8 a.m.	10
8 a.m. to 12 p.m.	16
12 p.m. to 4 p.m.	21
4 p.m. to 8 p.m.	18
8 p.m. to 12 a.m.	12

Decision Variables: $x_{\{8,t\}}$ = employees starting 8-hour shift at time t ; $x_{\{12,t\}}$ = employees starting 12-hour shift at time t . All ≥ 0 .

Objective: $\min Z = 744 * \sum x_{\{8,t\}} + 1209.6 * \sum x_{\{12,t\}}$ ($744 = 8 \times 93$, $1209.6 = 12 \times 100.8$)

Constraints (demand per period):

- T1 00–04: $x_{\{8,00\}} + x_{\{8,20\}} + x_{\{12,00\}} + x_{\{12,20\}} \geq 8$
- T2 04–08: $x_{\{8,00\}} + x_{\{8,04\}} + x_{\{12,00\}} + x_{\{12,20\}} \geq 10$
- T3 08–12: $x_{\{8,04\}} + x_{\{8,08\}} + x_{\{12,00\}} + x_{\{12,08\}} \geq 16$
- T4 12–16: $x_{\{8,08\}} + x_{\{8,12\}} + x_{\{12,08\}} + x_{\{12,12\}} \geq 21$
- T5 16–20: $x_{\{8,12\}} + x_{\{8,16\}} + x_{\{12,08\}} + x_{\{12,12\}} \geq 18$
- T6 20–24: $x_{\{8,16\}} + x_{\{8,20\}} + x_{\{12,12\}} + x_{\{12,20\}} \geq 12$

Optimal Solution (Gurobi):

- 8h shifts: 8@00, 2@04, 14@08, 6@12, 11@16, 0@20
- 12h shifts: 0@00, 0@08, 1@12, 0@20
- **Optimal daily cost:** \$31,713.60

```

8-hour shift starting at 00:00: 8.00 employees
8-hour shift starting at 04:00: 2.00 employees
8-hour shift starting at 08:00: 14.00 employees
8-hour shift starting at 12:00: 6.00 employees
8-hour shift starting at 16:00: 11.00 employees
8-hour shift starting at 20:00: 0.00 employees
12-hour shift starting at 00:00: 0.00 employees
12-hour shift starting at 08:00: 0.00 employees
12-hour shift starting at 12:00: 1.00 employees
12-hour shift starting at 20:00: 0.00 employees
Minimum total cost: $31713.60

```

Part 2: At most one-third of controllers can work 12-hour shifts

Add constraint: $\sum x_{12,t} \leq (1/3) * (\sum x_{8,t} + \sum x_{12,t})$. With the same optimal solution as Part 1, 12h staff = 1, total = 42, so $1/42 < 1/3$. The constraint is non-binding; optimal solution and cost remain the same: **\$31,713.60**.

Part 3: Staff the three airports separately

Model changes: (1) Index variables and constraints by airport: $x_{8,t}^{(i)}$, $x_{12,t}^{(i)}$ for airport $i = 1, 2, 3$; (2) Each airport has its own demand constraints per time period; (3) Objective = sum of costs across all three airports (still minimize total cost).

Additional data needed: Table 1 gives aggregate demand. To staff separately, we need **demand per airport per time period** $d_{i,t}$ (e.g., Airport A needs 3 in 00–04, Airport B needs 2, Airport C needs 3, summing to 8). Without this split, the separate-staffing model cannot be implemented.

Question 2

Primal Problem

```

max Z = 10*x_1 + 14*x_2 + 20*x_3
s.t.  2*x_1 + 3*x_2 + 4*x_3 <= 220
      4*x_1 + 2*x_2 - x_3   <= 385
      x_1 + 4*x_3           <= 160
      x_1, x_2, x_3 >= 0

```

Dual Problem

```

min W = 220*y_1 + 385*y_2 + 160*y_3
s.t.  2*y_1 + 4*y_2 + y_3   >= 10
      3*y_1 + 2*y_2         >= 14

```

$$4y_1 - y_2 + 4y_3 \geq 20$$

$$y_1, y_2, y_3 \geq 0$$

Question 3

Part 1: Sensible cutting patterns (leftover < 3 ft)

#	Pattern	3-ft	4-ft	5-ft	Used (ft)	Waste (ft)
1	3×3	3	0	0	9	1
2	2×4	0	2	0	8	2
3	2×5	0	0	2	10	0
4	2×3+1×4	2	1	0	10	0
5	1×3+1×5	1	0	1	8	2
6	1×4+1×5	0	1	1	9	1

Note: 1×3+1×4 gives leftover 3 ft, excluded by "leftover < 3 ft".

Part 2: Integer linear program

Variables: x_j = number of 10-ft boards cut with pattern j , $j = 1, \dots, 6$.

Objective: $\min \sum_{j=1}^6 x_j$

Constraints: For each length (3-ft, 4-ft, 5-ft), total yield $\geq$ demand:

- 3-ft: $3x_1 + 2x_4 + x_5 \geq 90$
- 4-ft: $2x_2 + x_4 + x_6 \geq 60$
- 5-ft: $2x_3 + x_5 + x_6 \geq 60$
- x_j in $\mathbb{Z}_+$ (nonnegative integers)

Optimal solution (Gurobi): Minimum **83** boards. Pattern 2: 8, Pattern 3: 30, Pattern 4: 45; others 0.

```
Minimum number of boards: 83
Pattern 2 (2×4): 8
Pattern 3 (2×5): 30
Pattern 4 (2×3+1×4): 45
```

Part 3: Minimize scrap (same board count)

Algebraic model: Let $N^* = 83$ be the minimum boards from Part 2. We solve a second ILP: $\min \sum_j w_j x_j$ s.t. $\sum_j x_j = N^*$, $\sum_j a_{jk} x_j \geq d_k$ for each length k , and x_j in $\mathbb{Z}_+$, where w_j = waste (ft) for pattern j , a_{jk} = yield of length k from pattern j , d_k = demand for length k . This keeps the board count at N^* while minimizing total scrap.

Optimal solution: Minimum scrap **14 ft**. Pattern 2: 7, Pattern 3: 30, Pattern 4: 46.

```
Minimum scrap (ft): 14
Pattern 2 (2×4): 7
Pattern 3 (2×5): 30
Pattern 4 (2×3+1×4): 46
```

Appendix: Code

staff_scheduling.py (Question 1)

```
#!/usr/bin/env python
# -*- coding: utf-8 -*-

from gurobipy import GRB, Model

demand = {
    "00:00-04:00": 8,
    "04:00-08:00": 10,
    "08:00-12:00": 16,
    "12:00-16:00": 21,
    "16:00-20:00": 18,
    "20:00-24:00": 12,
}

cost_8h_per_hour = 93
cost_12h_per_hour = 100.8
cost_8h_shift = cost_8h_per_hour * 8
cost_12h_shift = cost_12h_per_hour * 12

start_times_8h = ["00:00", "04:00", "08:00", "12:00", "16:00", "20:00"]
start_times_12h = ["00:00", "08:00", "12:00", "20:00"]

def solve_staffing_problem(apply_12h_shift_constraint=False):
    model = Model("Staffing_Problem")
    x_8h = model.addVars(start_times_8h, name="X_8h",
vtype=GRB.CONTINUOUS)
    x_12h = model.addVars(start_times_12h, name="X_12h",
vtype=GRB.CONTINUOUS)

    model.setObjective(
        sum(x_8h[t] * cost_8h_shift for t in start_times_8h) +
        sum(x_12h[t] * cost_12h_shift for t in start_times_12h),
        GRB.MINIMIZE
    )

    model.addConstr(x_8h["00:00"] + x_8h["20:00"] + x_12h["00:00"] +
x_12h["20:00"] >= demand["00:00-04:00"], "Demand_T1")
    model.addConstr(x_8h["00:00"] + x_8h["04:00"] + x_12h["00:00"] +
```

```

x_12h["20:00"] >= demand["04:00-08:00"], "Demand_T2")
    model.addConstr(x_8h["04:00"] + x_8h["08:00"] + x_12h["00:00"] +
x_12h["08:00"] >= demand["08:00-12:00"], "Demand_T3")
    model.addConstr(x_8h["08:00"] + x_8h["12:00"] + x_12h["08:00"] +
x_12h["12:00"] >= demand["12:00-16:00"], "Demand_T4")
    model.addConstr(x_8h["12:00"] + x_8h["16:00"] + x_12h["08:00"] +
x_12h["12:00"] >= demand["16:00-20:00"], "Demand_T5")
    model.addConstr(x_8h["16:00"] + x_8h["20:00"] + x_12h["12:00"] +
x_12h["20:00"] >= demand["20:00-24:00"], "Demand_T6")

    if apply_12h_shift_constraint:
        total_12h_staff = sum(x_12h[t] for t in start_times_12h)
        total_staff = sum(x_8h[t] for t in start_times_8h) +
total_12h_staff
        model.addConstr(total_12h_staff <= (1/3) * total_staff,
"12h_Shift_Limit")

    model.optimize()

    if model.status == GRB.OPTIMAL:
        for t in start_times_8h:
            print(f"8-hour shift starting at {t}: {x_8h[t].X:.2f}
employees")
        for t in start_times_12h:
            print(f"12-hour shift starting at {t}: {x_12h[t].X:.2f}
employees")
        print(f"Minimum total cost: ${model.ObjVal:.2f}")

if __name__ == "__main__":
    solve_staffing_problem(apply_12h_shift_constraint=False)
    solve_staffing_problem(apply_12h_shift_constraint=True)

```

lumberyard_cutting.py (Question 3)

```

#!/usr/bin/env python
# -*- coding: utf-8 -*-
"""IEOR 4004 Assignment 2 - Question 3: Lumberyard Cutting"""

from gurobipy import GRB, Model, quicksum

patterns = [
    {"name": "3x3", "yield": (3, 0, 0), "waste": 1},
    {"name": "2x4", "yield": (0, 2, 0), "waste": 2},
    {"name": "2x5", "yield": (0, 0, 2), "waste": 0},
    {"name": "2x3+1x4", "yield": (2, 1, 0), "waste": 0},
    {"name": "1x3+1x5", "yield": (1, 0, 1), "waste": 2},
    {"name": "1x4+1x5", "yield": (0, 1, 1), "waste": 1},
]
demand = {0: 90, 1: 60, 2: 60}

def solve_min_boards():

```

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    m = Model("lumberyard_min_boards")
    n = len(patterns)
    x = m.addVars(n, name="x", vtype=GRB.INTEGER, lb=0)
    m.setObjective(quicksum(x[j] for j in range(n)), GRB.MINIMIZE)
    for k in range(3):
        m.addConstr(quicksum(patterns[j]["yield"][k] * x[j] for j in
range(n)) >= demand[k])
    m.optimize()
    return m, x

def solve_min_waste(min_boards):
    m = Model("lumberyard_min_waste")
    n = len(patterns)
    x = m.addVars(n, name="x", vtype=GRB.INTEGER, lb=0)
    m.setObjective(quicksum(patterns[j]["waste"] * x[j] for j in
range(n)), GRB.MINIMIZE)
    m.addConstr(quicksum(x[j] for j in range(n)) == min_boards)
    for k in range(3):
        m.addConstr(quicksum(patterns[j]["yield"][k] * x[j] for j in
range(n)) >= demand[k])
    m.optimize()
    return m, x

if __name__ == "__main__":
    model1, x1 = solve_min_boards()
    if model1.status == GRB.OPTIMAL:
        min_boards = int(model1.ObjVal)
        print(f"Minimum number of boards: {min_boards}")
        for j in range(len(patterns)):
            if x1[j].X > 0:
                print(f"  Pattern {j+1} ({patterns[j]['name']}):
{x1[j].X:.0f}")
        model2, x2 = solve_min_waste(min_boards)
        if model2.status == GRB.OPTIMAL:
            print(f"Minimum scrap (ft): {model2.ObjVal:.0f}")
            for j in range(len(patterns)):
                if x2[j].X > 0:
                    print(f"  Pattern {j+1} ({patterns[j]['name']}):
{x2[j].X:.0f}")

```